



Department of Information Technology

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CD3291 & Data Structures and Algorithms

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV /AVL Trees
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Crossword Puzzle
- **Justification:**

In data structures, an AVL Tree is self-balancing binary search tree that maintains a strict balance condition to ensure efficient operations. AVL Trees provide fast and reliable access, insertion, and deletion operations by automatically adjusting the tree structure to maintain balance after every update. This property makes AVL Trees highly suitable for applications requiring quick lookups and dynamic data organization. The cross-word puzzle activity is used to check the understanding of students about the concept of AVL Trees.

- **Time Allotted for the Activity: 10 Minutes**
- **Details of the Implementation:**
 - In this activity the students were given with 10 crossword questions related to AVL Tree concept. Students needed to think critically and fill in the answers within the puzzle.
 - The activity took place 10 minutes in the class, followed by that a review session was conducted where the answers were checked. Sample solutions were displayed in Figure 1 & 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO4	3	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	2	2	2
Justification for correlation	The students apply the basic engineering knowledge to understand tree data structures.	The students analyze the given questions	The students find the solutions for the given question.	The students work in a team to find solutions for the given question.	The students communicate with their team members and explain the reason for providing particular solution for the question.	The students apply tree data structures to solve some computational problem.

• Images / Screenshot of the practice:

AVL TREE - CROSS WORD PUZZLE

M. Surulivelan
M. Mathan Raja.

ACROSS

- Which rotation is applied when the left subtree is too tall?
- The maximum allowed difference in heights of left and right subtrees is?
- What operation is performed to restore balance in an AVL tree?
- The rotation that fixes Left-Right imbalance is called?
- The height difference between left and right subtrees in an AVL tree node is called?

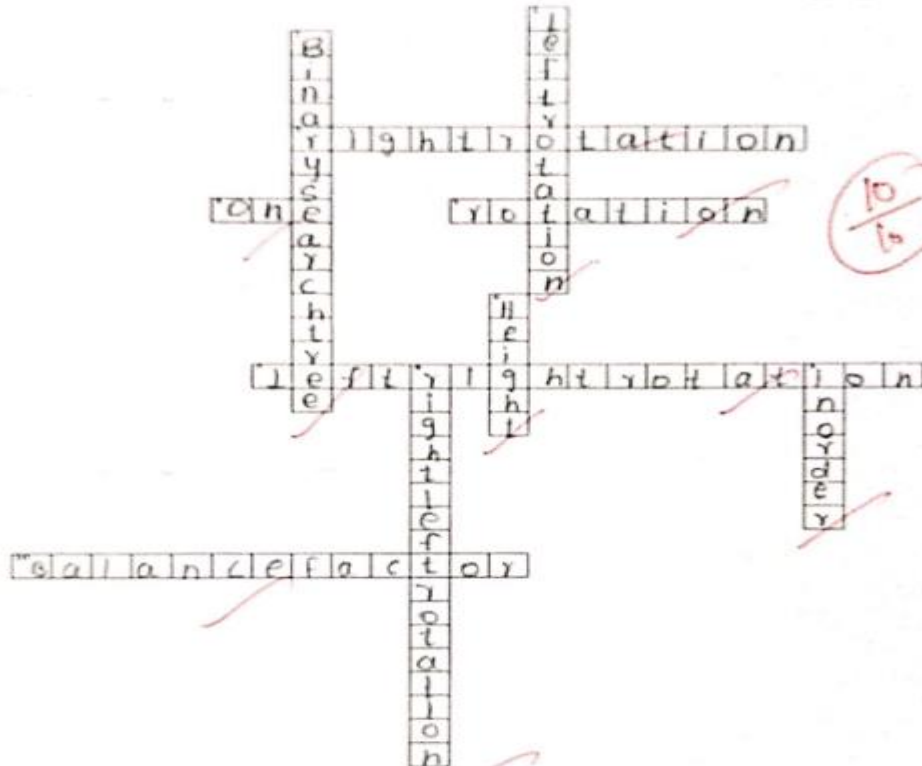
DOWN

- Which rotation is applied when the right subtree is too tall?
- AVL tree is a _____ type of tree
- After insertion, to check the balance, we update the ____ of nodes?
- The rotation that fixes Right-Left imbalance is called?
- The traversal technique used to get nodes in sorted order?

Figure 1: Sample Answers sheet shared by Surulivelan. M and Mathan Raja. M

AVL TREE - CROSS WORD PUZZLE

S Saranya
R. Devika



ACROSS

3. Which rotation is applied when the left subtree is too tall?
4. The maximum allowed difference in heights of left and right subtrees is?
5. What operation is performed to restore balance in an AVL tree?
7. The rotation that fixes Left-Right imbalance is called?
10. The height difference between left and right subtrees in an AVL tree node is called?

DOWN

1. Which rotation is applied when the right subtree is too tall?
2. AVL tree is a _____ type of tree
6. After insertion, to check the balance, we update the _____ of nodes?
8. The rotation that fixes Right-Left imbalance is called?
9. The traversal technique used to get nodes in sorted order?

Figure 2: Sample Answers sheet shared by Saranya. S and Devika. R

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Most of the students actively participated and enjoyed the challenge of solving the crossword puzzle.
- The activity helped to highlight areas where some students needed extra support or clarification.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - The activity helps reinforce vocabulary and terminology related to AVL Trees in an easy way.
 - It promotes critical thinking as students must connect clues to AVL Tree operations and characteristics.
 - They encourage students to recall key concepts like balancing, rotations, and tree properties.
- ❖ ***Challenges faced in implementation:***
 - Some students may find technical terms and concepts related to AVL Trees difficult to understand.
 - Students with less prior knowledge might struggle to solve the puzzle independently.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ <https://crosswordlabs.com/>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/2_GM_CS3353_Pointers_CWP.pdf
- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/GM_CS3491_U1_CWP.pdf

Signature of Faculty Member

HOD